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High Performance Processors and Systems

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Name
Last Name

Problem 1 (25%)	
Problem 2 (25%)	
Question 1 (25%)	
Question 2 (25%)	
Total (100%)	
Extra (3 points)	
Total (x% + 3 points)	

Problem 1

Assume that a given 5 stage pipeline MIPS architecture is working with a clock cycle equal to 2 ns. Using the MIPS code shown below,

I1: ADD \$s3, \$s2, 2
I2: ADD \$s5, \$s4, \$s3
I3: SW \$s5, 4(\$s3)
I4: SUB \$s7, \$s5, \$s6
I5: LW \$s6, 4(\$s7)

- Draw the pipeline schema showing all the RAW data conflicts. Solve the conflicts without using rescheduling and path forwarding.
- Reschedule the instructions to reduce the stalls; Draw the pipeline schema showing all the RAW data conflicts.
- The forwarding paths have been included in the pipeline. Draw the pipeline schema showing all the forwarding paths that have to be used to solve the conflicts.

Answer

- Show all the RAW data conflicts

Istr	CK1	CK2	CK3	CK4	CK5	CK6	CK7	CK8	CK9	CK10	CK11	CK12	CK13
I1	IF	ID	EX	MEM	WB								
I2		IF	ID	EX	MEM	WB							
I3			IF	ID	EX	MEM	WB						
I4				IF	ID	EX	MEM	WB					
I5					IF	ID	EX	MEM	WB				

CPU without forwarding

[illegible]

b) Rescheduled code:

CPU without forwarding

[illegible]

c) CPU with the forwarding paths

[illegible]

Problem 2

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.
(R0 is set to 1, R1 is set to 0)

```
LOOP:    LD      F3  0    R0
         ADDD    F1  F3   F3
         MULTD   F2  F3   F1
LOOP2:   LD      F3  0    R1
         MULTD   F2  F2   F3
         MULTD   F2  F2   F3
         BNEZ    R1  LOOP2
         SUBI    R0  R0   1
         BNE     R0  R1   LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? If no, please, change the value of R0 and R1 to make it. Please effectively support your answer.

Answer

Question 1

Define the main characteristics of a VLIW architecture. Explain which is the basic idea behind Trace Scheduling for VLIW and why is it necessary to get good performance from a VLIW?

Question 3

Define the performance measures, CPU time and MIPS. Why is CPU time a better performance measure than MIPS? How do you summarize performance when comparing between two alternatives over a set of benchmarks?

EXTRA

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
I1: LD F6 32+ R2	1	2	7	8
I2: ADDD F2 F6 F4	2	9	11	12
I3: MULTD F0 F4 F2	4	13	43	44
I4: SUBD F12 F2 F6	3	9	11	12
I5: ADDD F0 F12 F2	13	17	19	20

A. List all the possible conflicts in the code.

B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

C. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

Answer 1.A

Answer 1.B

Answer 1.C

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2				
LD F2 45+ R3				
MULTD F0 F4 F2				
ADD F2 F8 F6				
DIVD F12 F0 F6				
SUBD F6 F8 F2				